Weed Emergence Risk Mapping (WERM): A Log-Space Multiplicative Framework for Pre-Emergence Risk Estimation from Field Imagery and Thermal Accumulation

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Abstract

We present WERM, a framework estimating pre-emergence weed risk from RGB field imagery and thermal accumulation. Four drivers—soil moisture, residue cover, structural disturbance, and growing degree days—are fused via log-space weighted geometric aggregation with a residue-damped texture correction. On 333 field images, texture dominates spatial variance (94.5%) and is indispensable under ablation ($\rho = 0.652$ upon removal). Geometric fusion differs materially from an additive baseline ($\rho = 0.954$), with multiplicative suppression in low-moisture zones. Rankings are robust to weight perturbations and noise ($\rho \geq 0.990$). Independent ground truth (touch-ranked moisture $\rho = 0.879$; photo-grid residue) and spatial diagnostics on a 61-image geotagged grid (semivariogram ranges 35–46 m; significant clustering) confirm the framework captures ecologically plausible structure. High-risk zones correspond to areas of heaviest weed emergence in the preceding 2–3 seasons.

I. Introduction

Weed emergence remains a persistent agricultural challenge, with increasing herbicide resistance and shifting weed communities demanding a shift from reactive to predictive management [1–3]. Emergence timing and spatial heterogeneity strongly influence scouting allocation and pre-emergence herbicide decisions, motivating spatially resolved, time-specific approaches [4–7].

A practical gap exists between two mature research lines. Emergence forecasting models predict *when* weeds germinate using thermal or hydrothermal time, but are weakly coupled to meter-scale surface heterogeneity [4, 5]. Computer vision detects *where* weeds occur, but only post-emergence [8,9]. No existing system estimates *where* emergence risk is highest *before* weeds are visible.

Biologically, emergence requires simultaneous satisfaction of moisture, thermal, and physical-exposure conditions; a deficit in any single driver can suppress germination regardless of others [4, 10]. This motivates a multiplicative rather than additive formulation. Surface residue further complicates analysis by modifying both microclimate and image texture [10–12].

We propose WERM: a framework that estimates four surface-condition drivers from RGB imagery and meteorological data, fuses them via log-space weighted geometric aggregation with a residue-damped texture correction, and produces a 0–100 risk score with A–E operational bins. We validate through ablation, robustness analysis, expert agreement, independent ground-truth measurements, species-specific thermal sensitivity, additive-versus-geometric comparison, and geostatistical spatial diagnostics.

II. Related Work

Thermal and hydrothermal time models remain the strongest biological basis for emergence prediction [4–6], but are species-specific, calibration-dependent, and sensitive to burial conditions [13]. Published base temperatures vary by 1.5–2.7 °C across populations [18], and formal parameterization is unavailable for Indian ecotypes. This justifies retaining thermal accumulation as an ecophysiologically informed prior but not a sufficient standalone predictor. Crop residue alters microclimate and physical exposure [10, 11], and GLCM texture descriptors capture structural patterns [12]. These studies motivate integrating image-derived moisture, residue, and texture with thermal accumulation while explicitly modeling their interactions.

III. Methodology

A. Component Estimation

WERM estimates four drivers, each normalized to $[0, 1]$. The moisture proxy uses a ResNet-18 fine-tuned on the INSA Soil Moisture Regression Dataset (train MSE = 0.004). The residue fraction uses a ResNet-18 trained on the USGS residue cover dataset (validation MAE = 0.07). Structural disturbance is represented by global GLCM contrast (32 gray levels), percentile-normalized [12]. The thermal component uses

7-day GDD from NASA POWER API with default $T_b = 10\,^\circ$C, normalized by 120 °Cd. Table 1 summarizes the models.

B. Datasets

Primary dataset. 333 RGB field images acquired on 14 February 2026 at 22.783°N, 75.948°E across two wheat fields prior to visible emergence. GDD was constant (54.0 °Cd, $G_{\text{norm}} = 0.450$). Two farmers (20–30 years experience) assigned ordinal L/M/H labels for moisture, residue, and disturbance.

Ground-truth verification. Thirty soil samples with controlled water addition were touch-ranked driest-to-wettest by farmers and compared against CNN moisture predictions via Spearman correlation. Residue ground truth used the photo-grid method: 10×10 grid overlay on 30 images, 100-point classification per image, compared against CNN residue fractions.

Spatial verification. A separate 61-image geotagged dataset was collected on 18 March 2026 on a 7-row × 10-column grid (56.1 × 72.1 m, 3–7 m spacing) using NoteCam GPS (accuracy < 7 m).

C. Residue-Damped Texture and Fusion

Residue inflates image texture while suppressing emergence. The correction $T_{\text{eff}} = T_{\text{norm}}(1-R)^\alpha$ ($\alpha = 1$) decouples these effects. Risk is modeled as:

$$\text{risk} = M^{w_M} \cdot G^{w_G} \cdot (1-R)^{w_R} \cdot T_{\text{eff}}^{w_T} \qquad (1)$$

with w_M=0.30, w_G=0.20, w_R=0.20, w_T=0.30, ε=10^{-6}. In log space: $\log(\text{risk}) = \sum w_i \log(x_i)$, enabling elasticity interpretation. Scoring uses MAD-tanh compression to 0–100, partitioned into bins A–E. Table 2 lists all parameters. The additive baseline $R_{\text{add}} = \sum w_i x_i$ was compared using identical inputs and weights.

Table 2: WERM Fusion Parameters

Parameter	Value	Description
w_M (moisture)	0.30	CNN moisture proxy
w_G (GDD)	0.20	Normalized GDD
w_R (residue)	0.20	Uncovered fraction
w_T (texture)	0.30	Damped T_{eff}
ε	10^{-6}	Floor before log
Tillage floor	0.02	Smooth floor
γ	0.70	Compression
α	1.0	Damping exponent
T_b	10 °C	Default base temp.
GDD baseline	120 °Cd	Normalizing const.

D. Spatial Diagnostics

Coordinates were projected to meters. Empirical semivariograms were computed for moisture, residue, texture, and risk score (15 lags, max lag = half diagonal). Spherical models were fitted. Moran's I used distance-band weights (threshold = median semivariogram range, 42.3 m). Spatial block CV used 42.3 m cells.

IV. Results

A. Risk Distribution and Residue Damping

The MAD-tanh scaling pipeline produced a broad and operationally useful distribution of risk scores across the 333 field images without numerical collapse. The final class counts were A=119 (35.7%), B=37 (11.1%), C=24 (7.2%), D=38 (11.4%), and E=115 (34.5%), with mean 48.5, median 50.0, IQR [10.2, 87.1], and SD 37.7 (Figs. 1–2). The substantial separation between low- and high-risk zones, with intermediate classes preserved, supports the use of WERM as a relative field-risk ranking system.

Residue damping systematically reduced residue–texture coupling. The correlation $r(T_{\text{eff}}, R)$ changed from +0.155 at α=0 (undamped) to −0.406 at α=1 to −0.654 at α=2. Ranking stability remained high around the selected operating point: Spearman correlation relative to α=1 was 0.993 at α=1.5 and 0.982 at α=2 (Fig. 3). The selected α=1 balances contamination suppression with preservation of useful structural signal.

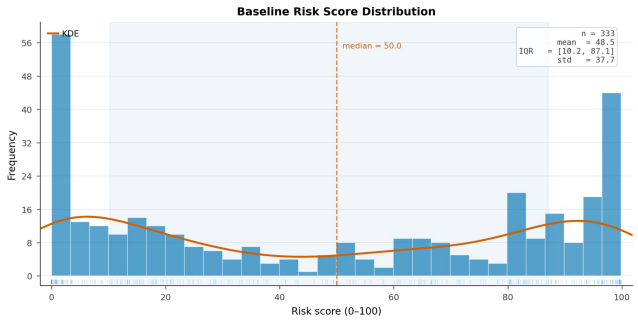


Figure 1: Baseline risk score histogram with KDE overlay ($n = 333$).

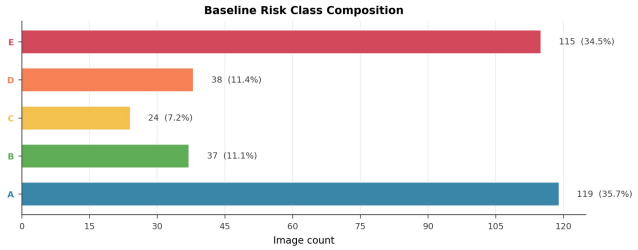


Figure 2: Risk class composition. A and E dominate; intermediate bins preserved.

Table 1: Summary of WERM Component Models

Component	Model	Dataset	Training	Output	Result
Moisture	ResNet-18	INSA Soil Moisture	Transfer + regression	$M \in (0, 1)$	MSE = 0.004
Residue	ResNet-18 + Sigmoid	USGS Residue Cover	Direct regression	$R \in [0, 1]$	MAE = 0.07
Structure	GLCM contrast	Field + ref. set	Non-learned	T_{norm}	Pct-norm.
Thermal	7-day GDD	NASA POWER API	Meteorological	$G \in (0, 1)$	Constant

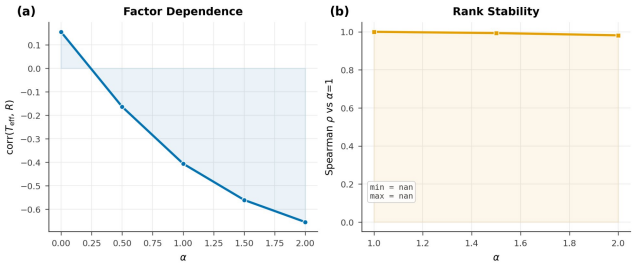


Figure 3: (a) $r(T_{\text{eff}}, R)$ across α. (b) Rank stability relative to α=1.



Figure 5: Factor contribution distributions. c_T widest spread (mean −0.496).

B. Variance Attribution

Log-space variance decomposition showed that structural disturbance was the dominant contributor to spatial variation. At α=1: Var(log risk) = 0.0746, with texture contributing Var(c_T) = 0.0633 (94.5%), residue 0.0033 (4.9%), moisture 0.00035 (0.5%), and GDD ≈ 0. The factor contribution violin plots (Fig. 5) confirm that the texture term exhibits the widest spread (mean −0.496) while moisture (−0.130) and GDD (−0.092) are tightly clustered. These results confirm that texture-derived disturbance is the primary spatial differentiator in this single-date dataset where GDD is constant (Figs. 4–5).

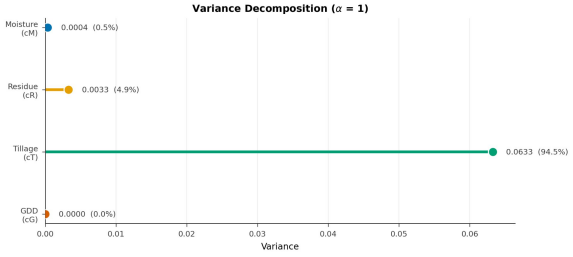


Figure 4: Log-space variance decomposition: texture 94.5%.

C. Additive Versus Geometric Fusion

The additive baseline and geometric fusion were related but not equivalent. Rankings showed Spearman $\rho = 0.954$ (Kendall $\tau = 0.839$), top-20% overlap 0.881, top-10% overlap 0.912, and mean absolute score difference of 5.39. The additive model assigned higher risk for 201/333 images, reflecting the systematic downward pull of the geometric model on images with at least one low input factor (Fig. 6).

Disagreement analysis revealed a clear pattern of multiplicative suppression. Of the 34 most rank-discrepant images (top 10% by absolute rank difference), 17 (50%) exhibited a "one-low-others-high" input profile—a single factor below 0.2 while remaining factors exceeded 0.6. The dominant suppressor was moisture: the most extreme case ($M = 0.026$, other inputs high) showed a geometric score of 5.5 versus an additive score of 46.9—a 41-point gap reflecting the geometric model's enforcement that emergence is negligible without adequate soil moisture (Fig. 7).

Expert agreement with farmer classifications was comparable between methods: geometric $\rho = 0.652$ (CI: 0.575–0.720), additive $\rho = 0.688$ (CI: 0.617–0.750), with overlapping confidence intervals precluding strong conclusions about which better captures expert judgment (Fig. 8). The spatial distribution of disagreement (Fig. 9) shows suppression zones are spatially distributed rather than concentrated in a single region, indicating the pattern reflects distributed moisture variation rather than a localized artifact.

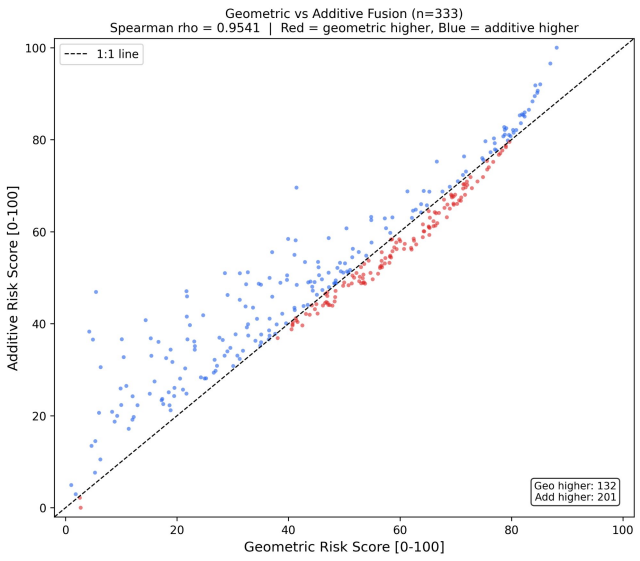


Figure 6: Geometric vs. additive scores. $\rho = 0.954$.

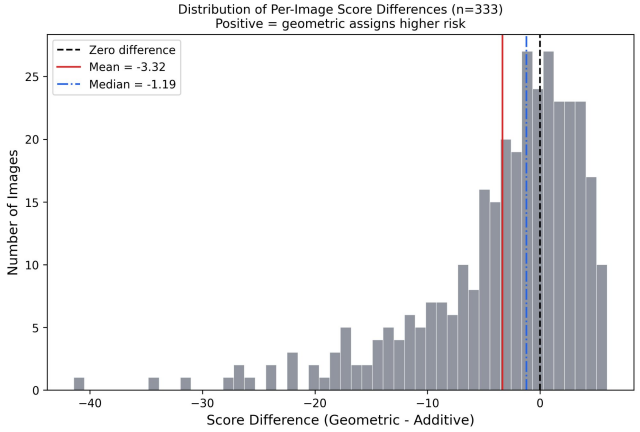


Figure 7: Score differences (geo − add). Mean = −3.32. Left tail: suppression.

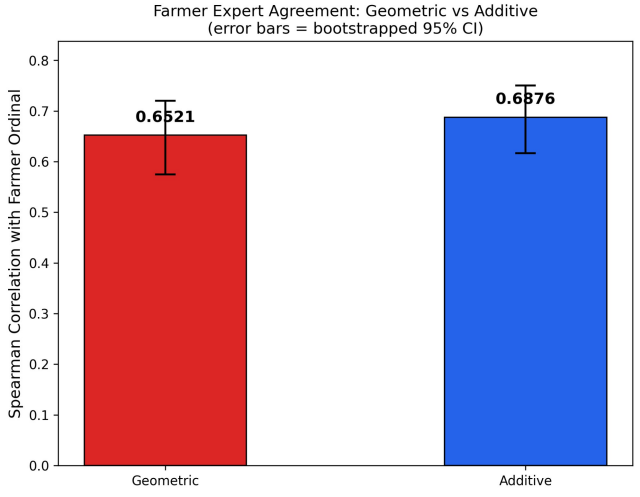


Figure 8: Expert agreement: geometric $\rho = 0.652$ vs. additive $\rho = 0.688$.

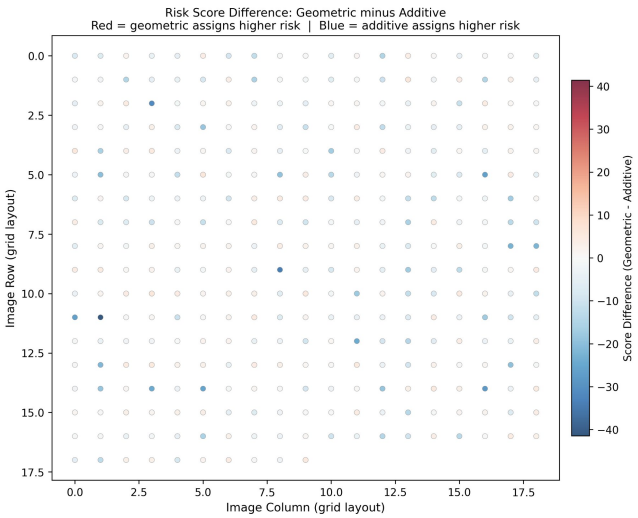


Figure 9: Spatial distribution of score differences. Suppression zones distributed.

D. Ablation, Robustness, and Weight Sensitivity

Leave-one-factor-out. Removing the disturbance term reduced Spearman to 0.652 and top-20% overlap to 0.375, indicating substantial collapse of rank structure. Removing residue had a moderate effect ($\rho = 0.968$, overlap 0.886), removing moisture was minimal ($\rho = 0.996$, overlap 1.000), and removing GDD produced no change ($\rho = 1.000$). These results are consistent with the variance analysis and confirm disturbance is the dominant spatial driver in this single-date dataset (Fig. 10).

Weight sweeps. Equal weights ($w_i = 0.25$): Spearman = 0.996, top-20% overlap = 0.970. Across 24 one-at-a-time sweeps (± 0.05, ± 0.10, ± 0.15 per factor): mean $\rho = 0.997$, min $\rho = 0.984$, min overlap = 0.859—both dips at texture -0.15, consistent with texture dominance. GDD sweeps produced $\rho = 1.000$ (Fig. 11).

Scaling. All alternative scalers preserved rank: z-tanh and min–max both $\rho = 1.000$, percentile clip $\rho = 0.9999$. Min–max distorted classes (D=178); MAD+tanh preserved balanced tails (Fig. 12). Prioritization is driven by the fusion model, not the score remapping.

Noise. At 1%/3%/5% multiplicative noise: $\rho = 0.999/0.996/0.990$; top-20% overlap = 1.000/0.970/0.913 (Table 3, Fig. 13). Even under 5% perturbation, high-risk prioritization is largely preserved. Epsilon floor: Spearman ≥ 0.9999 across $\varepsilon \in \{0.001, 0.05\}$.

Table 3: Robustness Summary

Test	Setting	Spearman	Top-20%
Leave-one-factor-out			
	−Moisture	0.996	1.000
	−Residue	0.968	0.886
	−Texture	0.652	0.375
	−GDD	1.000	1.000
Noise			
	$\sigma = 0.01$	0.999	1.000
	$\sigma = 0.03$	0.996	0.970
	$\sigma = 0.05$	0.990	0.913

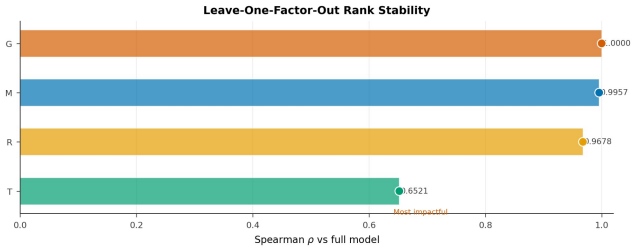


Figure 10: LOFO rank stability. Texture removal: strongest collapse.

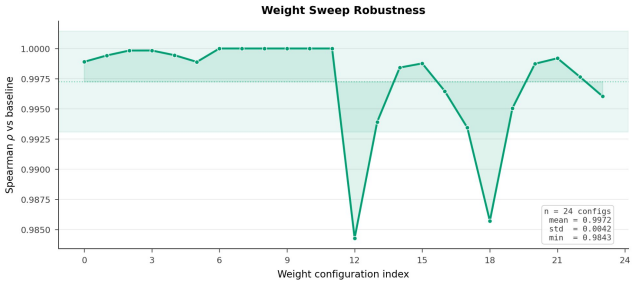


Figure 11: Weight sweep: mean $\rho = 0.997$, min $= 0.984$.

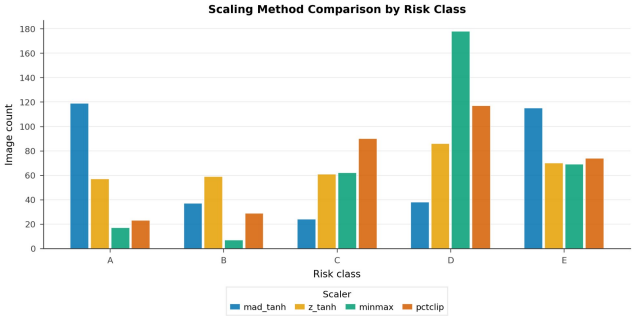


Figure 12: Scaler comparison. MAD+tanh preserves balanced tails.

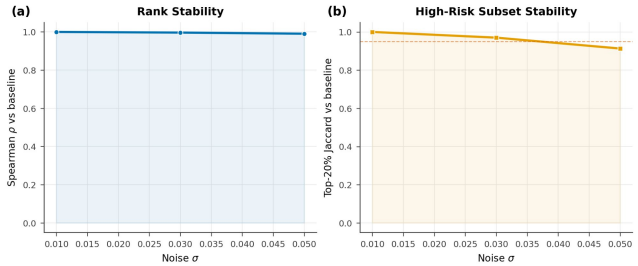


Figure 13: (a) Rank stability and (b) top-20% overlap under noise.

E. Dirichlet Weight-Ensemble Uncertainty

Dirichlet weight sampling (500 draws from $\mathrm{Dir}(1,1,1,1)$) showed that a stable high-risk core persists under broad weight uncertainty: mean $\rho = 0.945$ (5th pct: 0.775, min: 0.514), mean top-20% overlap $= 0.804$ (5th pct: 0.451). Despite this spread, 42 images remained in the top-20% set with probability ≥ 0.9, while only 26 exhibited high uncertainty (rank SD ≥ 50). This indicates uncertainty is concentrated in a limited subset of ambiguous cases rather than uniformly distributed (Figs. 14–15).

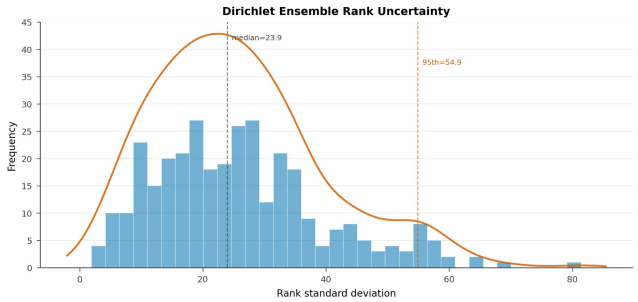


Figure 14: Rank SD under Dirichlet sampling. Median $= 23.9$, 95th pct $= 54.9$.

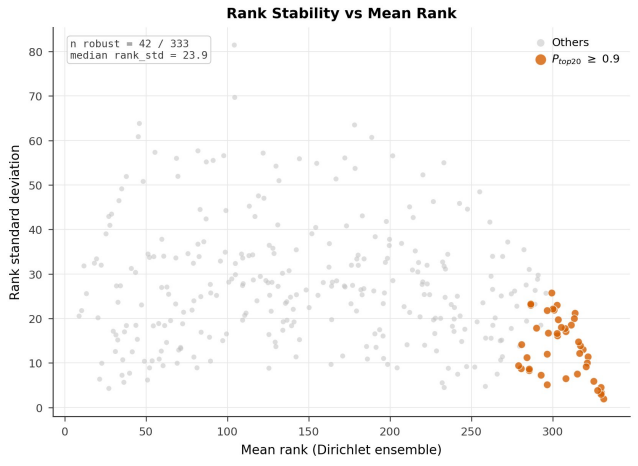


Figure 15: Mean rank vs. rank SD. Orange: 42 robust high-risk images ($P_{\text{top20}} \geq 0.9$).

F. Ground-Truth Verification

Component-level proxy validation showed strong agreement between model-derived ordinal bins and expert visual assessment: 97.8% for moisture, 93.7% for residue, and 96.5% for texture. Independent ground-truth verification strengthened this finding. The 30 touch-ranked moisture samples yielded Spearman $\rho = 0.879$ between farmer-ranked wetness ordering and CNN-predicted moisture ranking, confirming that the proxy reliably captures the moisture gradient. Photo-grid residue counting on 30 images (100-point grid overlay per image) confirmed that CNN residue fractions track visually counted residue cover.

Historical weed-cycle correspondence. Farmer consultation confirmed that WERM's identified mid-to-high risk zones (scores > 50) correspond spatially to areas where weed emergence was heaviest during the preceding 2–3 rabi growing seasons. While qualitative and retrospective, this spatial agreement provides preliminary external support for the framework's risk ranking.

G. Species-Specific Thermal Sensitivity

The default base temperature of $T_b = 10\,^{\circ}\mathrm{C}$ yielded GDD $= 54.0\,^{\circ}\mathrm{Cd}$ ($G_{\mathrm{norm}} = 0.450$). Species-specific recomputation produced substantially higher thermal accumulations (Table 4): *P. minor* ($T_b = 1.0\,^{\circ}\mathrm{C}$) yielded $125.5\,^{\circ}\mathrm{Cd}$ ($G_{\mathrm{norm}} = 1.000$), *C. album* ($T_b = 3.0\,^{\circ}\mathrm{C}$) yielded $111.5\,^{\circ}\mathrm{Cd}$ ($G_{\mathrm{norm}} = 0.929$), and *M. indica* ($T_b = 0\,^{\circ}\mathrm{C}$) yielded $132.5\,^{\circ}\mathrm{Cd}$ ($G_{\mathrm{norm}} = 1.000$). All three produced risk multipliers of 1.16–1.17×, corresponding to mean absolute score shifts of 12.6–13.8 points.

Because GDD is spatially constant, rank orderings are perfectly preserved under all three parameterizations ($\rho = 1.000$). The ecologically important finding is the magnitude of the shift: all three species show $G_{\mathrm{norm}} \geq 0.929$, indicating thermal conditions were already non-limiting at the collection date. Under species-specific parameterization, *P. minor* and *M. indica* reach GDD saturation, consistent with their known early emergence in Madhya Pradesh wheat during November–December. *C. album* accumulated slightly less, consistent with its later emergence peak (Figs. 16–17).

Table 4: Species-Specific Thermal Accumulation

Species	T_b	**GDD**	G_{norm}	**Risk**×	$\lvert\Delta\rvert$
Default	10.0°C	54.0	0.450	—	—
P. minor	1.0°C	125.5	1.000	1.17	13.8
C. album	3.0°C	111.5	0.929	1.16	12.6
M. indica	0.0°C	132.5	1.000	1.17	13.8

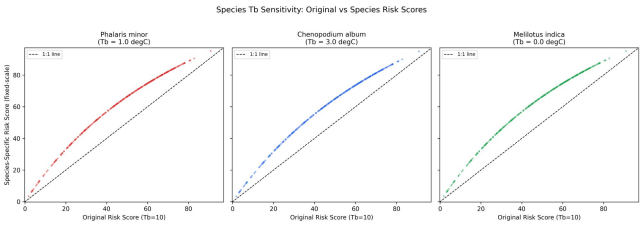


Figure 16: Species-specific vs. default scores. Perfect rank preservation.

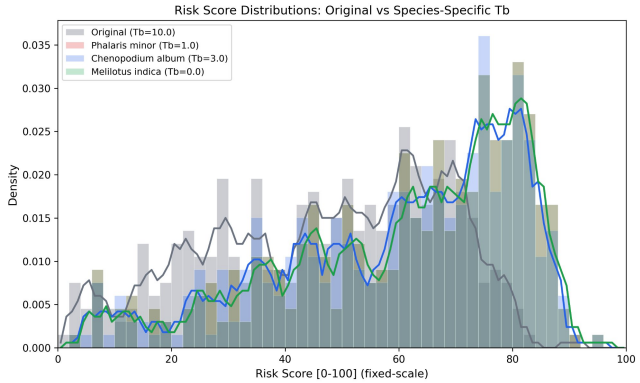


Figure 17: Score distributions shift rightward under species-specific T_b.

H. Spatial Diagnostics

Semivariograms fitted with spherical models revealed spatial correlation ranges of 38.8 m for moisture, 45.7 m for residue, 45.7 m for texture, and 35.2 m for the fused risk score (Table 5, Fig. 18). These ranges fall within the 5–50 m scale characteristic of weed-relevant surface patchiness, confirming ecologically plausible spatial structure rather than random noise.

Moran's I revealed significant positive spatial autocorrelation for residue ($I = 0.050$, $p = 0.001$) and texture ($I = 0.084$, $p = 0.001$), indicating spatially clustered surface properties consistent with tillage patterns. Moisture ($I = 0.021$, $p = 0.331$) and fused risk ($I = 0.011$, $p = 0.777$) were non-significant; the latter is attributable to the constant GDD term diluting per-component autocorrelation through the multiplicative fusion.

Spatial block cross-validation (42.3 m blocks) yielded three valid blocks with between-block variance accounting for 22.1% of total variance, confirming meaningful spatial risk zoning. Block mean risk scores were 38.7, 54.4, and 59.8, demonstrating distinct low, moderate, and high risk zones (Fig. 19). Mean within-block CV was 0.43, reflecting substantial local heterogeneity captured by the image-derived components.

Table 5: Spatial Diagnostics: Semivariograms and Moran's I

Variable	Range	Nugget	Sill	I	p
Moisture	38.8 m	0.0003	0.0006	0.021	0.331
Residue	45.7 m	0.0052	0.0244	0.050	0.001**
Texture	45.7 m	0.0026	0.1164	0.084	0.001**
Risk	35.2 m	383.6	585.4	0.011	0.777

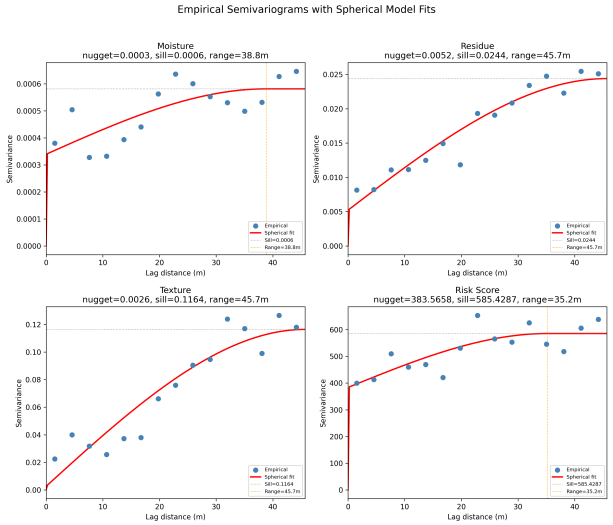


Figure 18: Semivariograms with spherical fits. Ranges 35–46 m.

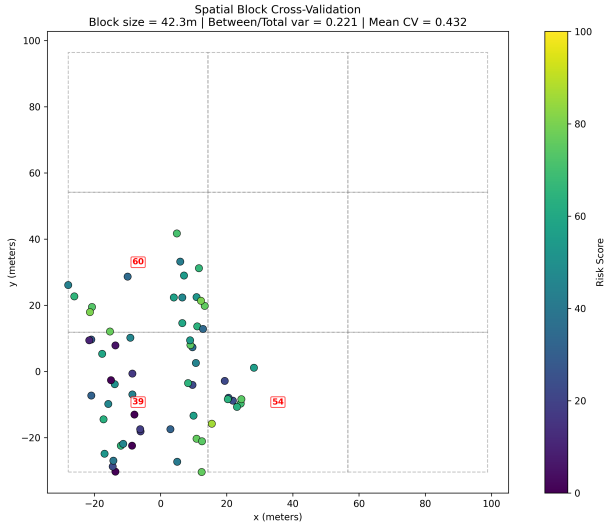


Figure 19: Spatial block CV. Block means: 38.7, 54.4, 59.8.

V. Discussion

Texture dominated spatial variance (94.5%) and was indispensable under ablation ($\rho = 0.652$), consistent with the single-date design where thermal accumulation was fixed. Residue retained a secondary role (4.9%); moisture's low spatial influence (0.5%) should increase in multi-date settings.

The multiplicative formulation is justified by the suppression analysis: half of the most rank-discrepant images showed one critically low factor with otherwise favorable conditions. The geometric model correctly suppresses risk in these cases, encoding the biological constraint that emergence requires simultaneous satisfaction of conditions [4, 6]. The slightly higher additive-expert agreement ($\rho = 0.688$ vs. 0.652) suggests farmer intuition may be partially compensatory—reasonable for scouting but potentially underweighting threshold effects. Resolving this definitively requires seasonal emergence data.

Species-specific analysis revealed that the default $T_b = 10\,^{\circ}\mathrm{C}$ substantially underestimates thermal accumulation. Published T_b values [14, 15, 17] indicate thermal conditions were already non-limiting at collection, consistent with observed phenology. Rank correlations remain perfect across all species (Fig. 20), confirming spatial risk invariance to T_b at any given date. Indian-ecotype values remain unavailable; given cross-origin variation of 1.5–2.7 °C [18], we plan a crop-aware automatic T_b pipeline: crop type + GPS + season → infer probable weed species → set T_b from published parameters.

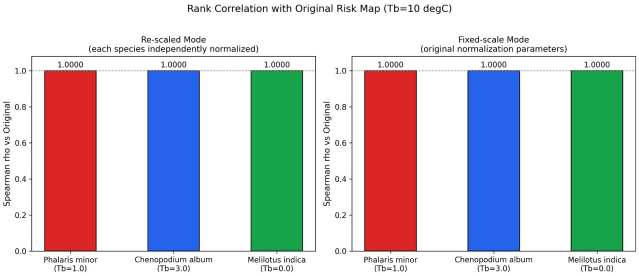


Figure 20: Rank correlation with original map: $\rho = 1.000$ for all species.

Ground-truth validation confirmed proxy accuracy at the rank level: touch-ranked moisture ($\rho = 0.879$) and photo-grid residue counting both supported the CNN outputs. The historical correspondence between WERM's high-risk zones and areas of heaviest weed emergence over 2–3 prior rabi seasons provides preliminary but meaningful external support.

Spatial diagnostics address the autocorrelation concern raised by Ploton et al. [19]. Semivariogram ranges of 35–46 m match the 5–50 m weed-patchiness scale. Significant clustering of residue and texture is ecologically expected; the non-significant Moran's I for fused risk reflects dilution by the constant GDD term, yet the interpolated risk map (Fig. 21) clearly shows spatial zoning with distinct hotspots—local pattern rather than broad gradient, consistent with weed emergence biology. Between-block variance of 22.1% confirms operationally meaningful risk differentiation.

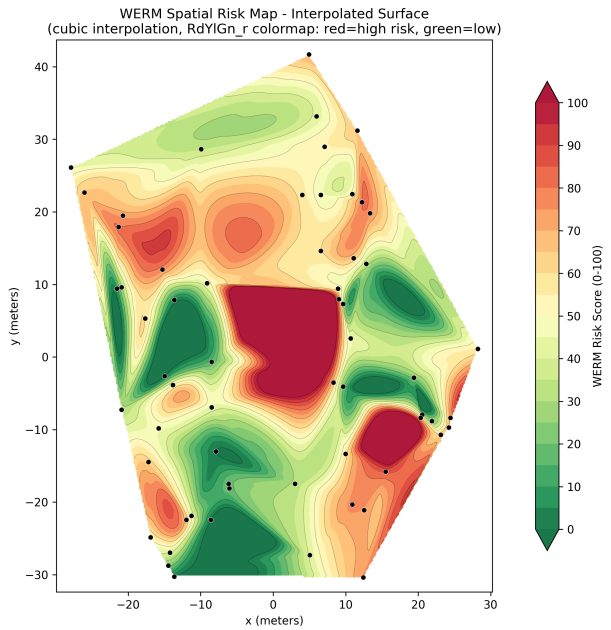


Figure 21: Interpolated WERM risk map. High-risk (red, 70–86) upper-left/center-right; low-risk (green, 0–30) lower-left.

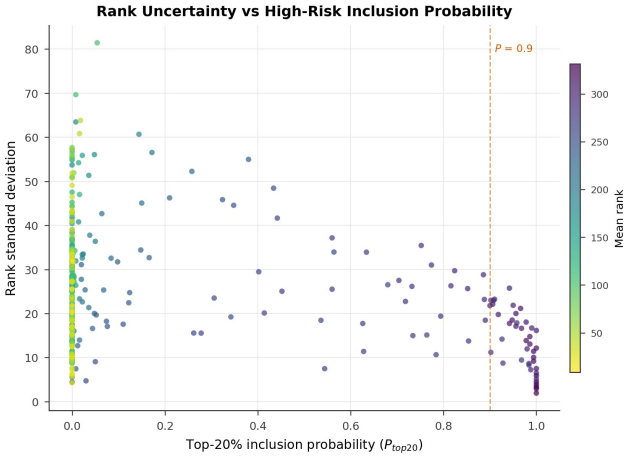


Figure 22: Rank uncertainty vs. top-20% probability. Rightmost cluster: robust scouting-priority zone.

VI. Limitations

The study uses single-date data without temporal dynamics. GDD was constant; species-specific analysis showed the default T_b underestimates accumulation. Touch-ranking and photo-grid methods confirm rank-level accuracy; absolute measurements (gravimetric moisture, TDR) would enable R^2/RMSE metrics. Historical weed-cycle correspondence is qualitative. Formal seasonal emergence validation with geolocated weed counts is the critical next step. A crop-aware T_b pipeline would transform WERM from a generic framework to a regionally calibrated system.

VII. Conclusion

WERM integrates image-derived moisture, residue, disturbance, and thermal accumulation via log-space multiplicative fusion. Texture dominates spatial variance (94.5%), geometric fusion correctly suppresses risk when single factors are low, and rankings are robust to perturbations. Expert agreement reached 93.7–97.8%, ground-truth validation confirmed proxy accuracy ($\rho = 0.879$ for moisture), and spatial diagnostics confirmed structure at weed-relevant scales (35–46 m). WERM's high-risk zones match historically observed weed-pressure patterns over 2–3 seasons. The framework establishes a coherent methodology for pre-emergence risk ranking, with seasonal emergence validation planned.

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